

From Semantic Mobility to Formal Representation

Mathematical Anchoring and Epistemic Provenance in Computational Knowledge Systems

Andy Cruccas
MARTRO Observatory

August 2026

Abstract

Modern AI systems can process large amounts of text, evidence and intermediate reasoning, yet the reliability of a computational result still depends on a prior problem: what exactly is being represented, under which meaning, and with which assumptions. Natural-language concepts are semantically mobile. Terms such as *distance*, *state*, *value*, *similarity*, and *risk* acquire different meanings across domains, while lexical similarity can invite analogies that are useful for discovery but invalid for inference. This paper proposes *mathematical anchoring* as an epistemic boundary between semantically mobile descriptions and deterministic computation. For a domain d , observations in an admissible domain X_d are mapped by an explicit formalization $\phi_d : X_d \rightarrow M_d$ into a declared mathematical structure M_d . A versioned projection $P_R : M_d \rightarrow \{0,1\}^n$ may then produce a finite computational state. The mathematical structure is not asserted to be the true ontology of the phenomenon; it is an operational model whose assumptions, validity scope, units, evidence bindings and provenance remain inspectable. This separation makes it possible to distinguish semantic suggestion from formal authority, discovery from validation, and computational determinism from ontological certainty. We analyze formalization as an information-selective operation, define the equivalence classes induced by successive representations, and introduce the idea of decision-relevant representational adequacy. SemaGraph-64 is presented only as a reference architecture: a six-bit finite-state terminal that demonstrates how extreme compression can remain auditable when the chain from evidence to formal structure to predicates is preserved. The broader contribution is therefore not a new finite-state tool, but a research program for epistemic systems engineering: designing human-machine systems that can generate candidate representations freely while preserving explicit mathematical contracts for authoritative computation.

Contents

1	Introduction: the representation problem	3
2	Semantic mobility and local ontology	3
2.1	Words do not carry fixed computational meaning	3
2.2	Relativity without total relativism	4
2.3	Analogy as hypothesis generation	4
3	A layered epistemic architecture	4
3.1	Phenomenon	5
3.2	Observation domain	5
3.3	Semantic interpretation	5
3.4	Mathematical structure	5
3.5	Computational representation	5

4	Mathematical anchoring	5
4.1	Definition	5
4.2	Candidate and registered anchors	6
4.3	Local stabilization	6
5	Formalization as information-selective compression	6
5.1	Equivalence classes induced by representation	6
5.2	Task-relative adequacy	7
5.3	Information gain is not knowledge gain	7
6	Provenance, replay and epistemic authority	7
6.1	Replayability	7
6.2	Historical non-rewriting	7
7	A small vocabulary of mathematical structure kinds	8
8	Discovery and validation in human-AI systems	8
8.1	Discovery mode	8
8.2	Validation mode	9
9	SemaGraph-64 as a reference architecture	9
9.1	Finite terminal	9
9.2	Trajectories rather than isolated states	9
9.3	Why six bits are not a universality claim	9
10	Worked conceptual examples	10
10.1	The word “distance”	10
10.2	Different words, same structure	10
10.3	Organizational feasibility	10
10.4	Probabilistic monitoring	10
11	From context engineering to epistemic systems engineering	10
12	Failure modes and limits	11
12.1	Wrong formalization, correct computation	11
12.2	False equivalence	11
12.3	Semantic leakage	11
12.4	Premature ontology selection	11
12.5	Formalism is not ontology	11
13	Research hypotheses and evaluation program	11
13.1	H1: explicit anchors reduce semantic drift	11
13.2	H2: provenance improves error localization	11
13.3	H3: decision-preserving compression can be measured	12
13.4	H4: semantic equality and structural equality diverge systematically	12
13.5	H5: ambiguity preservation outperforms forced anchor selection under sparse evidence	12
13.6	Evaluation dimensions	12
14	Relationship to established computational formalisms	12
15	Conclusion	13

A Reference implementation: computational notes	13
A.1 Transition surface	13
A.2 Branchless batch processing	14
A.3 Cross-implementation parity	14
B Reference implementation: trajectory identities	14

1 Introduction: the representation problem

The recent increase in computational capability has changed the bottleneck of many AI-assisted workflows. A contemporary system can generate, compare and revise large numbers of intermediate hypotheses, retrieve heterogeneous evidence, and execute many reasoning loops in parallel. The limiting question is therefore increasingly not whether a model can perform more computation, but whether the objects supplied to that computation have been represented in a way that preserves the distinctions relevant to the task.

This problem is older than large language models. Knowledge representation, formal logic, statistics, measurement theory and scientific modeling have always depended on the choice of what counts as an object, a relation and a valid transformation. What changes in AI systems is scale. A weak representational assumption that would once have produced one poor inference can now be propagated through hundreds or thousands of automated operations. Compute amplifies not only correct structure but also systematic representational error.

The central claim of this paper is narrow:

Before knowledge can be processed deterministically, the semantics used to describe a phenomenon must be locally stabilized by an explicit representational contract.

We call this operation *mathematical anchoring*. The purpose of an anchor is not to reveal the metaphysical essence of a phenomenon. Its purpose is to state, with enough precision to support computation, which mathematical structure is being used, which observations are admissible, which assumptions authorize the mapping, and which information is intentionally discarded.

The distinction matters because natural language is powerful partly because it is not rigid. A word can travel across domains and generate analogies. “Distance” can refer to Euclidean separation, shortest path in a graph, divergence between distributions, organizational separation, decision cost or social proximity. This semantic mobility is productive in discovery. It is dangerous in validation. Two relations can share a word without sharing mathematical properties, and two differently named relations can instantiate the same formal structure.

The proposed architecture therefore separates three operations that are often conflated:

1. **semantic interpretation**: generating candidate meanings and analogies;
2. **mathematical formalization**: selecting a declared structure under explicit assumptions;
3. **computational projection**: producing a deterministic state suitable for exact processing.

SemaGraph-64 is used later as a concrete finite-state terminal for this architecture. It is not the theory itself.

2 Semantic mobility and local ontology

2.1 Words do not carry fixed computational meaning

Natural-language terms are not reliable computational types. Their meaning is constrained by context, convention, use and the surrounding system of relations. The same lexical token may therefore refer to different mathematical objects.

Consider “distance.” In different domains it may mean:

- a metric $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ satisfying symmetry and the triangle inequality;
- shortest-path length on a graph;
- a directed divergence that is not symmetric;
- a cost-to-transform function;
- an ordinal judgment of separation with no cardinal interpretation.

The lexical label alone does not determine which operations are valid. If a system treats all occurrences of the word as equivalent, it commits an ontological error before any logical inference begins.

This gives a useful distinction.

Definition 1 (Inferential error). *An inferential error occurs when a conclusion does not follow from the stipulated premises and rules of the chosen representation.*

Definition 2 (Representational or ontological error). *A representational error occurs when the chosen entities, relations, scales or mathematical structure fail to represent the task-relevant phenomenon adequately, even if all subsequent inferences are formally valid.*

Mathematics is exceptionally strong against the first error once definitions and axioms are fixed. It does not by itself eliminate the second. A perfectly valid derivation can operate on the wrong model.

2.2 Relativity without total relativism

The claim that meaning is model-relative does not imply that truth is arbitrary. It implies that a proposition can only be evaluated after the interpretation under which its terms are used has been made explicit.

Let W denote a phenomenon in the world and let $M_1, M_2, \dots$ be alternative formal models. Different models may preserve different properties of W . Their adequacy can still be compared empirically or operationally by asking whether they preserve the relations needed for prediction, explanation, control or decision.

The relevant position is therefore neither naive realism about a single representation nor unrestricted relativism. It is *controlled representational pluralism*: multiple candidate formalizations may coexist, while every authoritative computation remains conditional on a declared model.

2.3 Analogy as hypothesis generation

Semantic mobility is not merely noise. It is one of the mechanisms by which cross-domain hypotheses can be generated. A relation in one domain can suggest a structurally similar relation in another. The important boundary is that analogy generates a candidate mapping; it does not validate it.

A disciplined cross-domain transfer has the form

$$D_A \rightarrow A^* \xrightarrow{\psi} B^* \rightarrow D_B, \quad (1)$$

where A^* and B^* are abstract structural descriptions and ψ is a proposed structure-preserving map. The scientific question is whether the relevant invariants survive the mapping. A verbal resemblance is insufficient.

3 A layered epistemic architecture

We distinguish five layers.

3.1 Phenomenon

The phenomenon is the portion of reality under investigation. It is not assumed to be fully observable or fully representable.

3.2 Observation domain

For domain d , let

$$X_d \tag{2}$$

denote the admissible observational domain: measurements, events, records, statements or other evidence accepted by the current protocol.

An element $x \in X_d$ is not yet a computational state. It is evidence represented in the vocabulary of the domain.

3.3 Semantic interpretation

Interpretation associates observations with one or more candidate conceptual structures. In human and LLM-assisted workflows this layer can remain flexible, probabilistic and plural. Competing interpretations should be allowed to coexist when evidence does not discriminate between them.

3.4 Mathematical structure

A mathematical anchor selects a declared structure

$$M_d \tag{3}$$

and a formalization map

$$\phi_d : X_d \rightarrow M_d. \tag{4}$$

The map may be partial: not every observation must be admissible, and insufficient evidence may block formalization.

3.5 Computational representation

A computational projection maps the formal representation to an object designed for deterministic processing. In the finite-state case,

$$P_R : M_d \rightarrow \{0, 1\}^n, \tag{5}$$

where R denotes a registered projection version. The composed anchor is

$$A_{d,R} = P_R \circ \phi_d : X_d \rightarrow \{0, 1\}^n. \tag{6}$$

This chain should never be collapsed conceptually:

$$\text{phenomenon} \neq \text{observation} \neq \text{mathematical model} \neq \text{computational state}. \tag{7}$$

4 Mathematical anchoring

4.1 Definition

Definition 3 (Mathematical anchor). *A mathematical anchor is a versioned contract that specifies a formalization map from an admissible observation domain into a declared mathematical structure together with the assumptions, validity scope, units, evidence bindings and provenance required to interpret that map.*

A minimal anchor contract contains:

- domain identifier and version;
- mathematical structure kind;
- variables and relations;
- observation bindings;
- assumptions;
- validity scope;
- dimensional or unit constraints where applicable;
- projection definition and version;
- evidence and provenance requirements.

The contract establishes computational authority, not ontological truth.

4.2 Candidate and registered anchors

A central design decision is to separate candidate generation from authoritative registration.

Definition 4 (Candidate anchor). *A candidate anchor is a proposed formalization generated by a human, language model or automated discovery process. It may be plausible, useful or elegant, but it is non-authoritative.*

Definition 5 (Registered anchor). *A registered anchor is a candidate that has been made explicit, versioned, structurally validated and accepted under a declared governance process. Only a registered anchor may produce authoritative deterministic states.*

This distinction allows exploratory reasoning to remain creative without allowing semantic speculation to mutate canonical computational state.

4.3 Local stabilization

Mathematical anchoring is a form of local semantic stabilization. It does not make the underlying word universally precise. It states instead:

For this task, under these assumptions and within this validity scope, the relevant relation will be treated as an instance of this mathematical structure.

The word remains mobile outside the contract. Computation becomes rigid inside it.

5 Formalization as information-selective compression

Formalization is not a neutral transcription. It is an information-selective operation.

5.1 Equivalence classes induced by representation

Every map $\phi_d : X_d \rightarrow M_d$ induces an equivalence relation

$$x \sim_\phi x' \iff \phi_d(x) = \phi_d(x'). \quad (8)$$

Observations that are distinct in X_d but identical under ϕ_d become computationally indistinguishable at the mathematical-model layer.

Likewise, a projection $P_R : M_d \rightarrow \{0, 1\}^n$ induces

$$m \sim_P m' \iff P_R(m) = P_R(m'). \quad (9)$$

The complete pipeline therefore creates nested equivalence classes:

$$X_d \xrightarrow{\phi_d} M_d \xrightarrow{P_R} \{0, 1\}^n. \quad (10)$$

The engineering question is not whether information is lost. Any useful compression loses information. The question is whether the lost distinctions matter for the task.

5.2 Task-relative adequacy

Let $\mathcal{T}$ denote a set of downstream queries, predictions or decisions. A representation is adequate relative to $\mathcal{T}$ if distinctions that can change the correct result for tasks in $\mathcal{T}$ are preserved with sufficient reliability.

This suggests a task-relative notion of distortion. Let $L_{\mathcal{T}}(x, r)$ denote the loss incurred when representation r is used in place of observation x for the declared task set. A representation design problem can then be written schematically as

$$\min_R C(R) + \lambda \mathbb{E}[L_{\mathcal{T}}(X, R(X))], \quad (11)$$

where $C(R)$ is representational cost. This is a research formulation, not a claim that one universal loss function exists.

The important shift is conceptual: representational compression should be evaluated by what it preserves for inference and action, not by compactness alone.

5.3 Information gain is not knowledge gain

Shannon information measures statistical surprise under a coding distribution [1]. Epistemic value is different. A random bit string may have high Shannon information while contributing almost nothing to a decision. Conversely, a short causal or structural constraint can carry enormous decision value.

A mature epistemic system must therefore distinguish at least:

1. statistical information;
2. semantic interpretation;
3. epistemic relevance to belief revision;
4. decision relevance to action.

Mathematical anchoring operates between the second and third layers: it converts a semantic candidate into a structure on which explicit inference can be licensed.

6 Provenance, replay and epistemic authority

A finite state is trustworthy only if its derivation remains inspectable. The authoritative lineage should support traversal in both directions:

$$\text{state bit} \rightarrow \text{predicate} \rightarrow \text{formal variable/relation} \quad (12)$$

$$\rightarrow \text{mathematical anchor} \rightarrow \text{observation binding} \quad (13)$$

$$\rightarrow \text{evidence}. \quad (14)$$

This makes two properties possible.

6.1 Replayability

Given identical evidence, anchor version and projection version, the formalized object and computational state should be reproducible.

6.2 Historical non-rewriting

A later anchor version may reinterpret the same evidence, but it must not silently rewrite the state produced by an earlier registered version. Historical results belong to the contract under which they were generated.

This is particularly important in agentic systems. Without versioned lineage, a model can revise an assumption and unknowingly make earlier reasoning appear to have been based on the new assumption. Provenance converts such drift into an explicit model transition.

7 A small vocabulary of mathematical structure kinds

The purpose of a mathematical-domain registry is not to create a universal ontology of mathematics. It is to provide explicit types for recurrent structural commitments.

Useful initial families include:

Family	Example structure	Typical question licensed by the structure
Metric/ geometric	metric, pseudometric, norm	How far apart are two objects under declared axioms?
Order	partial order, total order, ranking	Is one state before, below, preferred to or dominated by another?
Graph	directed graph, path, connectivity	Which dependencies, reachability relations or bottlenecks exist?
Probability/ information	distribution, conditional probability, entropy, divergence	How is uncertainty or dependence represented under a declared probabilistic model?
Dynamical	state space, transition relation, discrete dynamical system	How does the system move under a transition rule?
Constraint	feasible set, boundary, admissible region	Which configurations are allowed or viable?
Decision/value	utility, cost, objective, preference relation	Which comparisons are licensed by the chosen evaluative structure?
Similarity	similarity function, equivalence relation, kernel-like relation	Which objects may be grouped or compared, and under which invariances?
Uncertainty	probability, intervals, bounded sets	What form of unresolved variation is being represented?

A lexical term does not determine one family. “Value” may be a scalar measurement, utility, price, ranking, or objective functional. “State” may be a point in a phase space, an aggregate label, a belief state or a software state. The anchor must declare which one is operative.

8 Discovery and validation in human-AI systems

Large language models are unusually strong at the semantically permissive part of reasoning: association, analogy, cross-domain retrieval, candidate synthesis and reformulation. Those properties are useful precisely because they are not confined to one formal ontology.

The same flexibility becomes dangerous when it is allowed to establish computational authority automatically. An LLM can propose that a phenomenon is well represented by a metric space, graph, partial order or probability model. It should not be able to make that representation authoritative merely by proposing it.

The architecture therefore separates two modes.

8.1 Discovery mode

Discovery may be broad and plural. The system may produce multiple candidate anchors, explain analogies, identify possible invariants and search for structure-preserving mappings across domains.

8.2 Validation mode

Validation is restrictive. A candidate must declare assumptions, domains and codomains, required evidence, mathematical properties and failure conditions. Only registered definitions enter the authoritative computational path.

This separation is the computational analogue of a familiar scientific distinction: intuition and analogy may generate hypotheses; proof, measurement and experiment determine which claims are retained.

9 SemaGraph-64 as a reference architecture

SemaGraph-64 provides a deliberately small terminal for testing these principles. Its role is not to model a domain completely. Its role is to demonstrate that a highly compressed deterministic state can remain epistemically auditable when the upstream anchor is explicit.

9.1 Finite terminal

Let

$$Q_6 = \{0, 1\}^6, \quad |Q_6| = 64. \quad (15)$$

A registered projection $P_R : M_d \rightarrow Q_6$ evaluates six deterministic predicates over the formalized representation. A state therefore records six declared binary distinctions, not six universal properties of the world.

For states $s, t \in Q_6$, the transition mutation is

$$m(s, t) = s \oplus t, \quad (16)$$

and Hamming distance is

$$d(s, t) = \text{popcount}(s \oplus t). \quad (17)$$

The complete direct transition basis contains $64 \times 64 = 4096$ ordered pairs.

9.2 Trajectories rather than isolated states

The system retains ordered trajectories because endpoint equivalence does not imply path equivalence. For a chain $C = (s_0, \dots, s_n)$, the ordered mutation signature preserves the sequence of changes, while net mutation

$$M_{\text{net}} = s_0 \oplus s_n \quad (18)$$

discards order and is therefore explicitly treated as a lossy projection.

This distinction is epistemically important beyond the implementation: a compressed summary must not be promoted into an identity criterion when distinct histories can produce the same summary.

9.3 Why six bits are not a universality claim

The Q_6 terminal is intentionally small. It is useful as an experimental surface because compression is extreme and false equivalence is easy to study. Nothing in the framework claims that six bits are sufficient to represent arbitrary phenomena. The number of bits, predicate design and mathematical anchor are task-dependent design choices.

SemaGraph therefore demonstrates a boundary condition: severe computational compression can be acceptable if the discarded distinctions are explicit, raw evidence is retained, and downstream claims remain conditional on the registered representation.

10 Worked conceptual examples

10.1 The word “distance”

Suppose three domains use the same word.

Physical position. A measurement system reports Cartesian coordinates. The anchor declares a Euclidean space and a metric. The triangle inequality is licensed.

Dependency network. Two components are separated by three directed edges. The anchor declares a graph and shortest-path relation. Symmetry may fail.

Decision cost. Moving from policy a to policy b has a transition cost. The reverse transition may have a different cost. Calling the quantity “distance” does not make it a metric.

A language model may notice the analogy across all three. Mathematical anchoring prevents the analogy from silently importing invalid operations.

10.2 Different words, same structure

Conversely, “precedence”, “dominance” and “dependency order” may instantiate partial orders in different domains. A structural comparison can therefore be meaningful even when the surface vocabulary differs. What matters is not shared terminology but preservation of the formal relations declared relevant.

10.3 Organizational feasibility

An organizational review may describe possible actions in terms of approvals, skills, budget and dependencies. Rather than treating “readiness” as one scalar concept, a candidate anchor can model the problem as a feasibility relation over constraints. Six registered predicates may then ask whether selected capability boundaries are satisfied. The resulting State64 vector is not the organization. It is a finite projection of one explicit feasibility model.

10.4 Probabilistic monitoring

A monitoring system may represent an observed signal by a probability distribution and information relations. A State64 projection may encode whether entropy, concentration or selected conditional relations cross declared thresholds. Again the state is authoritative only relative to the probability model and threshold contract that generated it.

11 From context engineering to epistemic systems engineering

Much contemporary AI practice focuses on context assembly: selecting documents, memories and tool outputs that should be placed in front of a model. This is necessary but not sufficient. A context can contain all relevant evidence while still failing to distinguish observation from hypothesis, semantic label from mathematical type, or current fact from superseded interpretation.

Knowledge engineering asks a deeper question: what structure should the cognitive asset have independently of a single prompt? Mathematical anchoring adds another layer: when does a semantic representation become sufficiently explicit to authorize deterministic inference?

The complete pipeline can be expressed as

$$\text{reality} \rightarrow \text{observation} \rightarrow \text{semantic hypotheses} \tag{19}$$

$$\rightarrow \text{mathematical anchors} \rightarrow \text{computational states} \tag{20}$$

$$\rightarrow \text{inference} \rightarrow \text{decision} \rightarrow \text{new observation.} \tag{21}$$

This suggests a broader research area that may be called *epistemic systems engineering*: the engineering of human-machine systems that acquire, represent, formalize, compare, revise and operationalize knowledge while preserving the provenance of the assumptions that make computation possible.

In such systems, knowledge engineering, decision engineering and organizational architecture become coupled. Knowledge determines what is represented; decision architecture determines which transformations from belief to action are permitted; organizational architecture determines where knowledge and decision rights reside across human and artificial agents.

12 Failure modes and limits

12.1 Wrong formalization, correct computation

The most important failure mode is an unjustified ϕ_d . A deterministic system can process a bad representation flawlessly. Registration and replay establish explicit authority, not truth.

12.2 False equivalence

Compression merges distinct objects. The system must identify which equivalence classes are created and prevent lossy summaries from being treated as full identity when the task requires distinctions they discard.

12.3 Semantic leakage

Labels attached for explanation can gradually acquire authority they were never intended to have. Mathematical structure kinds, predicate definitions and evidence bindings must remain the authoritative contract.

12.4 Premature ontology selection

When several anchors are plausible and evidence is insufficient, ambiguity should remain explicit. Automatic selection merely converts epistemic uncertainty into hidden system state.

12.5 Formalism is not ontology

A successful model may support prediction or decision without establishing that its objects correspond to the ultimate ontology of the phenomenon. The framework is deliberately agnostic on that metaphysical question.

13 Research hypotheses and evaluation program

The framework generates empirical and engineering questions that can fail.

13.1 H1: explicit anchors reduce semantic drift

Agentic systems using registered mathematical anchors should exhibit fewer representation-induced inconsistencies across long reasoning chains than systems relying on natural-language definitions alone, holding evidence constant.

13.2 H2: provenance improves error localization

When an output is challenged, systems retaining bit-to-predicate-to-anchor-to-evidence lineage should localize the source of disagreement more reliably than systems retaining only prompts and final states.

13.3 H3: decision-preserving compression can be measured

For a declared task set $\mathcal{T}$, finite projections can be evaluated by the rate at which they preserve downstream rankings or decisions relative to richer representations.

13.4 H4: semantic equality and structural equality diverge systematically

Benchmarks built around polysemous terms should demonstrate that lexical equality frequently fails to imply mathematical-anchor equality, while structurally equivalent relations can occur under different terminology.

13.5 H5: ambiguity preservation outperforms forced anchor selection under sparse evidence

When observations underdetermine the formalization, systems that preserve multiple candidate anchors should be better calibrated than systems that force one ontology prematurely.

13.6 Evaluation dimensions

A research program should therefore measure:

- replay consistency;
- provenance completeness;
- anchor-version stability;
- semantic-collision error rate;
- task-relative distortion after projection;
- ranking or decision preservation;
- ambiguity calibration;
- update cost when anchors change.

14 Relationship to established computational formalisms

The proposal does not replace existing formal systems. It specifies a boundary around their use.

Tradition	Relevant contribution	Distinction in this paper
Information theory	Quantification of uncertainty and coding [1]	Statistical information alone does not determine semantic or epistemic authority.
Error-correcting codes	Finite symbolic structure and Hamming geometry [2]	Used as computational mathematics, not as a theory of meaning.
Automata theory	Finite states and transitions [3]	The focus here is the upstream authorization of a state representation, not language recognition.
Statecharts / timed automata	Structured state-transition modeling [4, 5]	Mathematical anchoring concerns how domain observations become formal states before transition logic is applied.

AI knowledge representation	Explicit representation and inference [8]	The present emphasis is on plural semantic candidates, registered mathematical authority and provenance across the representation boundary.
Non-ergodic / regime-switching models	Path and regime dependence [6, 7]	These are possible domain formalisms; they do not define the anchoring framework itself.

15 Conclusion

The central problem addressed here is not computational scarcity but representational authority. Natural language is semantically mobile, and that mobility is productive for discovery. Deterministic computation requires a different property: locally fixed meaning under an explicit formal contract.

Mathematical anchoring provides such a boundary. An observation domain X_d is mapped into a declared mathematical structure M_d by a versioned formalization ϕ_d . A registered projection may then map that structure into a computational state. The mathematical model is neither hidden nor universalized: assumptions, validity scope, evidence and provenance remain part of the authority chain.

This architecture permits a useful division of labour. Humans and language models may generate analogies and competing ontologies freely. Authoritative computation begins only after a representation has been made explicit enough to be tested, versioned and replayed. The resulting system can therefore be semantically flexible upstream and computationally rigid downstream.

SemaGraph-64 demonstrates one extreme case: a six-bit terminal whose state-transition operations are fully deterministic while the richer domain meaning remains outside the kernel. Its significance is not that the world can be reduced to 64 states, but that severe compression can remain epistemically honest when the transformations that produced it are visible.

The broader research program is consequently larger than SemaGraph itself. It concerns the engineering of systems capable of moving from ambiguous language to explicit mathematical structure to deterministic action without confusing a useful representation with reality. That problem sits at the intersection of knowledge engineering, information theory, AI systems, decision science and the philosophy of scientific modeling, and may become increasingly central as computational reasoning ceases to be the scarce resource.

A Reference implementation: computational notes

This appendix records implementation details for SemaGraph-64. They are intentionally separated from the theoretical contribution.

A.1 Transition surface

A six-bit state occupies the low six bits of a byte. The ordered transition index is

$$I(s, t) = 64s + t \in [0, 4096). \quad (22)$$

Per-transition metadata can be derived from the mutation mask and stored in compile-time lookup tables. The current implementation uses approximately 4.4 KiB for its principal classification surface, allowing the hot path to remain cache-resident on conventional CPUs.

A.2 Branchless batch processing

The implementation provides a scalar table path and a struct-of-arrays batch API designed for vectorization. An optional explicit SIMD path is required to reproduce the scalar result exactly. These are engineering properties, not theoretical claims about the representation framework.

A.3 Cross-implementation parity

The TypeScript layer acts as executable specification and the Rust core as the machine-near implementation. Exhaustive fixtures verify the full 4096 transition basis, while seeded trajectory fixtures verify shape, dwell and projection parity. The purpose is to ensure that implementation changes cannot alter canonical structural outputs silently.

B Reference implementation: trajectory identities

For trajectories over the eight-symbol alphabet $Q_3 = \{0, \dots, 7\}$, SemaGraph distinguishes raw path identity, run-collapsed shape and dwell duration. Adjacent duplicate states are collapsed into runs, while non-adjacent returns remain distinct. The pair (shape, dwell) reconstructs the original path losslessly. Coarser quantities such as endpoint and net mutation are treated as pre-filters only.

This rule embodies the general epistemic principle developed in the paper: a lossy projection must not silently become an identity criterion.

References

- [1] C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, 1948.
- [2] R. W. Hamming, “Error Detecting and Error Correcting Codes,” *Bell System Technical Journal*, 1950.
- [3] J. E. Hopcroft, R. Motwani, and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*. Pearson, 2006.
- [4] D. Harel, “Statecharts: A Visual Formalism for Complex Systems,” *Science of Computer Programming*, 1987.
- [5] R. Alur and D. L. Dill, “A Theory of Timed Automata,” *Theoretical Computer Science*, 1994.
- [6] O. Peters, “The ergodicity problem in economics,” *Nature Physics*, 2019.
- [7] J. D. Hamilton, “A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle,” *Econometrica*, 1989.
- [8] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*. Pearson, 2021.